Homework 4

Computational Bootcamp, Summer 2026

Due: Wednesday, August 26

Question 1: Aiyagari done right (45 points)

Households have CRRA preferences ($u(c) = c^{1-\gamma}/(1-\gamma)$, $\gamma = 2$, $\beta = 0.96$) and cannot borrow, $a' \geq 0$. Income is an employment shock, $y \in \{y_e, 0\}$, on the Markov chain $\Pi = \left[\begin{smallmatrix} 0.95 & 0.05 \\ 0.75 & 0.25 \end{smallmatrix}\right]$: a 5% separation rate and a 75% job-finding rate, so $u = 6.25\%$, and $y_e = 1/(1-u) = 1.0667$ makes $E[y] = L = 1$. The household's problem is the same as Huggett (HW3 Q3), but with wage w:

$$V(a, y) = \max_{a' \geq 0} \left\{ u(c) + \beta \sum_{y'} \Pi(y' \mid y) V(a', y') \right\}, \qquad c = (1+r)a + wy - a'.$$

Prices come from a firm producing $Y = K^\alpha L^{1-\alpha}$, $\alpha = 0.36$, $\delta = 0.08$, whose first-order conditions give, for any r,

$$\frac{K}{L} = \left(\frac{\alpha}{r+\delta}\right)^{\frac{1}{1-\alpha}}, \qquad w(r) = (1-\alpha)\left(\frac{K}{L}\right)^\alpha.$$

With $L = 1$, that $K/L = K$ is capital demand. Note that because $y = 0$ in the low state, $a \geq 0$ is the natural borrowing limit and doesn't bind in equilibrium: a household at $a = 0$ who draws unemployment would consume zero, which Inada conditions forbid.

Code structure. Organize the model as: Parameters & solutions structs `AiyagariParameters`, `AiyagariSolutions` holding the policies, distribution, μ, the prices r and w, and aggregates Y, K, and `initialize(; kwargs...)` returning `(para, sols)`. Operators take `(para, sols)` and return the next iterate; solvers loop until convergence, write into `sols`, and return the iteration count.

A. Write the EGM operator `egm_step(para, sols)` and the solver `solve_egm!(para, sols)`, which iterates it to a fixed point in c and stores the policies in `sols`. You can reuse your code from HW3 Q3 but ensure it is adapted to this context. Also write `capital_demand(r, para)` and `wage(r, para)` from the firm's first-order conditions. Use a capital grid with $a_{\max} \approx 50$; guard against $c \leq 0$ at $a_{\min}$ when unemployed, or you risk $u'(c) = \infty$.

B. Solve for the stationary distribution using two methods: `simulate_panel!(para, sols, y_panel)` ($N = 10{,}000$ on the interpolated policy, on an income panel `simulate_y_panel(para)` drawn once; report your burn-in), and the histogram method — `transition_matrix(para, sols)` with the Young lottery (mass $\omega = (a_{k+1} - a')/(a_{k+1} - a_k)$ on a_k, the rest on a_{k+1}) and `stationary_distribution!(para, sols)` iterating $\mu' = (T^*)'\mu$. Wrap one as `aggregate_savings(r, para, sols)` and plot $A(r)$ at $\approx$ 15 rates under *each* method. Report mean assets, the bottom-50% and top-10% wealth shares, and the mass at both ends of the grid. For the shares write `top_wealth_share(a, μ, p)`, the fraction of total wealth held by the richest p of households. Why is the fraction of households at the borrowing constraint zero here?

C. Let `excess_demand(r, para, sols)` = `aggregate_savings(r, para, sols)` − `capital_demand(r, para)`. Solve for equilibrium r by root finder (`clear_market!(para, sols)`, reasonable bounds $r \in [-0.01, \frac{1}{\beta} - 1]$) and by the damped update $K_{n+1} = (1-\lambda)K_n + \lambda A(r(K_n))$ (`clear_market_damped!(para, sols; λ = 0.5)`); have both return (r^*, K^*) and the solved `(para, sols)`. Check that $r^* < 1/\beta - 1 = 4.17\%$ and that K^* exceeds its complete-markets value $(\alpha/(1/\beta - 1 + \delta))^{1/(1-\alpha)}$: that gap is precautionary saving.

D. Change the income risk: $p_{ue} : 0.75 \to 0.50$, renormalizing $y_e = 1/(1-u)$ so mean income is unchanged — passed in through `kwargs...`. Report the new (r^*, K^*) and explain briefly why a representative-agent model cannot produce this comparative static.

Question 2: A multinomial logit (25 points)

`data/choices.csv` records $N = 2{,}000$ consumers, each buying one of $J = 4$ products or nothing (the outside option, $j = 0$). It has one row per consumer-product pair, with columns `consumer`, `product`, `x`, `price`, `chosen` — where `chosen` is 1 on the row the consumer picked, so someone who bought nothing has four zeros. Consumer i values product j at $u_{ij} = \beta x_j - \alpha p_{ij} + \varepsilon_{ij}$ and the outside option at $u_{i0} = \varepsilon_{i0}$, with ε Type-I extreme value, x_j product j's characteristic, and p_{ij} the price consumer i faces for it, so

$$P_{ij}(\theta) = \frac{\exp(\beta x_j - \alpha p_{ij})}{1 + \sum_{k=1}^{J} \exp(\beta x_k - \alpha p_{ik})}, \qquad \theta = (\beta, \alpha).$$

The data were simulated at $\theta = (1.5, 1.0)$, so you can check your answer against the truth.

A. Read in the file and store the data as a DataFrame. Report the mean price and the share of consumers choosing each product, and the share buying nothing. Then write `load_choices(path)`, returning x (length J), the $N \times J$ matrix of prices, and the chosen index $j_i \in \{0, 1, \ldots, J\}$ — sorting by `consumer` then `product` makes the reshape easier.

B. Write `choice_probs(θ, p)`, returning the $J+1$ probabilities for one consumer (outside option first), and `neg_loglik(θ, data)` $= -\sum_i \ln P_{ij_i}(\theta)$, computed by subtracting $\max_k \delta_{ik}$ rather than exponentiating and dividing. Minimize it with Optim in `estimate_mlogit(data)`. Note $\alpha > 0$ and the optimizer does not know that, so estimate $a = \log \alpha$ and exponentiate inside. Do you recover θ?

C. Report standard errors: invert the numerical Hessian of $-\log L$ at $\hat{\theta}$, check that it is positive definite, and convert $\mathrm{se}(\hat{a})$ into $\mathrm{se}(\hat{\alpha})$ with the delta method. Is the truth within two standard errors?

D. Average the predicted probabilities at $\hat{\theta}$ across consumers and compare them to the shares you computed in part A. Then raise every consumer's price of product 4 by 10% and recompute: report the percentage change in each option's share, the outside option included. Where do product 4's lost buyers go? That pattern is IIA — are there products for which it would be a bad assumption?

Question 3: SMM on your own Aiyagari model (30 points)

Estimate $\theta = (\beta, \gamma)$ to match two moments computed from `data/hw4_wealth.csv`, a cross-section of $N = 5{,}000$ household wealth holdings drawn from the model at a θ_{true} we are not telling you:

the capital-output ratio K/Y and the top-10% wealth share.

A. Write `data_moments(path)`, returning $K/Y = \bar{a}^{1-\alpha}$ (mean wealth $\bar{a}$, since $Y = K^\alpha$ with $L = 1$), the top-10%, and the bottom-50% wealth shares (the last for part C) — reuse `top_wealth_share`. Write `model_moments(`θ`; shocks = nothing, kwargs...)` which solves the Question 1 equilibrium at θ and returns the same three moments; `smm_objective(`θ`, m_data, W; shocks = nothing, kwargs...)` returns $(m_{data}-m(\theta))'W(m_{data}-m(\theta))$. θ passes straight through to `initialize` via `kwargs...`, so Question 1's code is reused unchanged. Parts A–B match only the first two moments ($W = I$, 2×2); you can use a log reparameterization to enforce $\gamma > 0$. Either stationary-distribution method from Question 1 works here: the histogram is deterministic and ignores `shocks`, but if you use the simulated panel, draw the uniforms once outside and pass them in through `shocks`, so that $m(\theta)$ is a fixed function of θ rather than a fresh random draw at every trial θ.

B. Estimate with Nelder–Mead (`estimate_smm(m_data, W; shocks = nothing, kwargs...)`); report $\hat{\theta}$, the number of model solves, and wall time. Keep each solve cheap: a coarse asset grid, the histogram rather than the panel, and a warm start on r (keep one `sols` alive across solves). Then difference $\partial m/\partial\theta'$ (four numbers) and read it as an identification argument.

C. Add the bottom-50% wealth share as a third moment (already returned by `data_moments` and `model_moments`) and re-run `estimate_smm` with all three, $W = I$. Two parameters, three moments: report $\hat{\theta}$ and the minimized objective, and confirm it is not exactly zero. Which moment fits worst, and why can't (β, γ) reconcile it with the other two?

D. Re-estimate against the empirical targets $K/Y \approx 3$ and a US top-10% wealth share of ≈ 0.76. You will not match them: report what the optimizer does to (β, γ) as it tries, and what is missing from a model whose only heterogeneity is a two-state employment shock.

Question 4 (optional, ungraded): AI practice, with receipts

Redo **one** part of Questions 1–3 with an AI tool. In half a page: your prompts, one thing the tool got wrong and how you caught it (or, if it was flawless, the test that convinced you), and where you will not trust these tools without verification.

Question 5 (optional, ungraded): Your own model

Express an economic problem that interests *you* as a dynamic programming problem, solve it, and report what you find: decision rules, simulated outcomes, comparative statics. This is the whole course in miniature, and a fine way to start a second-year paper.